

is reduced to (3, 64, 64) in latent space. As a result, it requires 64 times less memory. This is why it is possible to generate high-resolution images quickly, even on GPUs with limited memory like the 16GB Colab GPUs.

Stable Diffusion during inference

The stable diffusion model receives a latent seed and a text prompt as inputs. The latent seed generates random latent image representations of size 64×64 , while the text prompt is transformed into text embeddings of size 77×768 using CLIP's text encoder.

The U-Net then denoises the random latent image representations iteratively while being conditioned on the text embeddings. The U-Net's output, which is the noise residual, is used to compute a denoised latent image representation through a scheduler algorithm. Several scheduler algorithms can be used for this computation, each with its own advantages and disadvantages. The recommended schedulers for Stable Diffusion are the PNDM scheduler (default), DDIM scheduler, and K-LMS scheduler.

The notebook does not delve into the details of how the scheduler algorithm works. In summary, the scheduler algorithm calculates the predicted denoised image representation based on the previous noise representation and the predicted noise residual. For more information, readers are encouraged to refer to the paper "Elucidating the Design Space of Diffusion-Based Generative Models."

The denoising process is iterated about 50 times to gradually obtain better latent image representations. Once completed, the latent image representation is decoded using the decoder portion of the variational autoencoder.

How to write your own inference pipeline?

We will now demonstrate how to customize diffusion pipelines using diffusers. This advanced feature can be particularly useful for replacing specific components, such as the VAE or scheduler described earlier.

The pre-trained model contains all the necessary components for setting up a complete diffusion pipeline. These components are stored in separate folders:

1. **Text_encoder:** Stable Diffusion utilizes CLIP, while other diffusion models might use other encoders like BERT.
2. **Tokenizer:** It must match the one used by the text_encoder model.
3. **Scheduler:** The scheduling algorithm that adds noise progressively to the image during training.
4. **UNet:** The model that generates the latent representation of the input.